COMPUTER NETWORKS

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ORANGE LEVEL PROBLEM

Name: Tanisha Mathur SRN: PES1UG24CS491 Section: I

Terminal 1

```
kavisha@keys: ~/sdn_project

[sudo] password for kavisha:
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...

*** Orange Network Started. Run 'pingall' to see stats. ***
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> iperf h1 h2
*** Iperf: testing TCP bandwidth between h1 and h2
*** Results: ['27.5 Gbits/sec', '27.4 Gbits/sec']
mininet> h1 ping -c 10 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=10.3 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.772 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=24.9 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.081 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.072 ms
64 bytes from 10.0.0.3: icmp_seq=6 ttl=64 time=0.052 ms
64 bytes from 10.0.0.3: icmp_seq=7 ttl=64 time=0.123 ms
64 bytes from 10.0.0.3: icmp_seq=8 ttl=64 time=0.087 ms
64 bytes from 10.0.0.3: icmp_seq=9 ttl=64 time=0.073 ms
64 bytes from 10.0.0.3: icmp_seq=10 ttl=64 time=0.114 ms

--- 10.0.0.3 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9139ms
rtt min/avg/max/mdev = 0.052/3.653/24.903/7.697 ms
```

Terminal 2

```
kavisha@keys: ~/sdn_project/

kavisha@keys:~/sdn_project/pox$ python3 pox.py orange_monitor
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.10.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:orange_monitor:Switch 00-00-00-00-00-01 connected. Monitoring logic active.
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:       1       42        2 06:87:f2:8e:78:78
INFO:orange_monitor:       1       42        3 06:87:f2:8e:78:78
INFO:orange_monitor:       1       42        3 06:84:00:49:ab:74
INFO:orange_monitor:       1       42        2 06:87:f2:8e:78:78
INFO:orange_monitor:       1       42        3 06:84:00:49:ab:74
INFO:orange_monitor:       1       42        3 06:87:f2:8e:78:78
INFO:orange_monitor:       1       42        1 06:84:00:49:ab:74
INFO:orange_monitor:       1       42        2 7a:83:ef:f0:36:eb
INFO:orange_monitor:       1       42        1 7a:83:ef:f0:36:eb
INFO:orange_monitor:       1       98        1 06:84:00:49:ab:74
INFO:orange_monitor:       1       98        2 06:87:f2:8e:78:78
INFO:orange_monitor:       1       98        1 7a:83:ef:f0:36:eb
INFO:orange_monitor:       1       98        3 06:87:f2:8e:78:78
INFO:orange_monitor:       1       98        2 06:87:f2:8e:78:78
INFO:orange_monitor:       1       98        1 06:84:00:49:ab:74
INFO:orange_monitor:       1       98        2 7a:83:ef:f0:36:eb
INFO:orange_monitor:       1       98        3 06:84:00:49:ab:74
INFO:orange_monitor:       1       98        3 06:87:f2:8e:78:78
INFO:orange_monitor:       1       98        1 7a:83:ef:f0:36:eb
INFO:orange_monitor:       1       98        3 06:84:00:49:ab:74
INFO:orange_monitor:       1       98        2 7a:83:ef:f0:36:eb
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
```

```
kavisha@keys: ~/sdn_project/

INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:       5      338        1 06:84:00:49:ab:74
INFO:orange_monitor:       1       66        2 06:87:f2:8e:78:78
INFO:orange_monitor:       1       54        2 06:87:f2:8e:78:78
INFO:orange_monitor:  387487 17199106286        1 06:84:00:49:ab:74
INFO:orange_monitor:  198109 13075366        2 06:87:f2:8e:78:78
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:       7      686        1 7a:83:ef:f0:36:eb
INFO:orange_monitor:       7      686        3 06:87:f2:8e:78:78
INFO:orange_monitor:       2       84        3 06:87:f2:8e:78:78
INFO:orange_monitor:       2       84        1 7a:83:ef:f0:36:eb
INFO:orange_monitor:       1       42        1 7a:83:ef:f0:36:eb
INFO:orange_monitor:       1       42        3 06:87:f2:8e:78:78
INFO:orange_monitor:--- Statistics Request Sent to Switches ---
INFO:orange_monitor:
[STATS REPORT] Switch ID: 00-00-00-00-00-01
INFO:orange_monitor:PACKETS  BYTES    PORT_IN  ETH_DST
INFO:orange_monitor:-------- -------- -------- -----------------
INFO:orange_monitor:      10      980        1 7a:83:ef:f0:36:eb
INFO:orange_monitor:      10      980        3 06:87:f2:8e:78:78
```

Terminal 3

```
kavisha@keys: ~

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kvsha>ubuntu
kavisha@keys:~$ sudo ovs-ofctl dump-flows s1
[sudo] password for kavisha:
kavisha@keys:~$ sudo ovs-ofctl dump-flows s1
 cookie=0x0, duration=8.251s, table=0, n_packets=9, n_bytes=882, idle_timeout=10, hard_timeout=30, priority=65535,icmp,i
n_port="s1-eth1",vlan_tci=0x0000,dl_src=26:53:7b:ee:c6:42,dl_dst=82:ba:8e:75:e8:a8,nw_src=10.0.0.1,nw_dst=10.0.0.3,nw_to
s=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth3"
 cookie=0x0, duration=8.249s, table=0, n_packets=9, n_bytes=882, idle_timeout=10, hard_timeout=30, priority=65535,icmp,i
n_port="s1-eth3",vlan_tci=0x0000,dl_src=82:ba:8e:75:e8:a8,dl_dst=26:53:7b:ee:c6:42,nw_src=10.0.0.3,nw_dst=10.0.0.1,nw_to
s=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth1"
 cookie=0x0, duration=3.021s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_
port="s1-eth1",vlan_tci=0x0000,dl_src=26:53:7b:ee:c6:42,dl_dst=82:ba:8e:75:e8:a8,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,arp_o
p=1 actions=output:"s1-eth3"
 cookie=0x0, duration=3.018s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_
port="s1-eth3",vlan_tci=0x0000,dl_src=82:ba:8e:75:e8:a8,dl_dst=26:53:7b:ee:c6:42,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,arp_o
p=2 actions=output:"s1-eth1"
kavisha@keys:~$
```

1. What is my project?

My project is an SDN-based Network Simulation designed to demonstrate the separation of the Control Plane and the Data Plane. It implements a "Traffic Monitoring and Statistics Collector" that provides real-time visibility into network flows, satisfying the "Orange Problem" requirements.

2. What do the two files do?

topology.py (The Data Plane Builder)

This file defines the physical (virtual) layout of your network.

- Function: It creates the hosts and the switch and connects them with virtual Ethernet (veth) links.
- SDN Role: It tells Mininet to use a Remote Controller, meaning the switch will not make any forwarding decisions on its own; it will wait for instructions from 127.0.0.1:6633.

orange_monitor.py (The Control Plane Logic)

This is the "Brain" of your network.

- Packet Handling: It handles packet_in events. When the switch sees a new packet it doesn't recognize, it sends it to this script.
- Flow Rule Design: The script tells the switch to "learn" the MAC addresses and install a Match-Action rule so future packets are handled at wire speed without asking the controller again.
- Monitoring: It uses a Timer to periodically (every 10 seconds) send ofp_stats_request messages to the switch to retrieve and log packet and byte counts.

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4. SDN Network Layer Implementation

- Application Layer: orange_monitor.py script acts as the application.
- Control Layer (Control Plane): The POX Controller uses the OpenFlow Protocol to communicate with the switch.
- Infrastructure Layer (Data Plane): The Open vSwitch (s1) simply forwards packets based on the flow table entries provided by the controller.

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5. Why didn't i use a Router? Why use a Switch?

- Centralized Intelligence: In a traditional network, you need a router to handle complex logic (like IP routing between subnets). In SDN, the "intelligence" is moved to the Controller. A switch is used because the Controller can program it to act like a switch, a router, or a firewall simply by changing the flow rules.
- Flow-Based Forwarding: Switches are more efficient for Match-Action logic. You used a switch because OpenFlow is designed to manipulate flow tables (Layer 2 MACs or Layer 3 IPs) within the switch hardware/software.
- Simplified Data Plane: Using a switch keeps the Data Plane simple and inexpensive, which is one of the core benefits of SDN—moving the complexity to a programmable software layer rather than expensive, fixed-function hardware routers.

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6. SDN Concepts

- Northbound Interface: The API between your script (orange_monitor.py) and the POX Controller.
- Southbound Interface: The OpenFlow protocol used between the POX Controller and the Open vSwitch.
- Flow Table: The database inside the switch that stores the "Match" (e.g., Destination MAC) and the "Action" (e.g., Output to Port 2).